

Here is how music travels from your phone screen into your ears, explained in simple, everyday steps:

[Tap Play on Spotify]

|



1. Phone Reads Digital File (Computers only know 1s & 0s)

|



2. Phone's Sound Card Translates (Turns 1s & 0s into real electrical sound waves)

|



3. Electric Signals Travel Down the Cable (Left & Right sound streams through the wire)

|



4. Earphone Speaker Vibrates (Tiny magnet & membrane shake back and forth)

|



5. Air Vibrations Hit Your Eardrum —▶ 🎵 You hear music!

Step 1: You Press Play on Spotify

Inside Spotify, a song is stored as computer data—a giant list of **numbers (1s and 0s)**, like a written recipe for sound. Your phone processor reads these numbers at lightning speed.

Step 2: The Phone Translates Numbers into Sound

Your ears cannot understand computer numbers, only sound waves.

- Inside your phone is a tiny **sound translator (called a DAC)**.
- It takes those 1s and 0s and instantly converts them into a smooth **electrical current** that pulses and waves just like the singer's voice and the instruments.

Step 3: Traveling Down the Cable

- When you plug in your Type-C audio cable, the phone opens a path for that electrical wave.
- The wave splits into two paths: one for your **Left ear** and one for your **Right ear**.
- It travels through the copper wires inside the cable straight to your earphones.

Step 4: Making the Sound in the Earphones

Inside each earphone bud is a tiny speaker made of:

1. A small **magnet**
2. A tiny **wire coil**
3. A very thin **plastic membrane (diaphragm)**

When the electrical wave reaches the coil, it turns it into a temporary magnet that pushes and pulls against the permanent magnet. This causes the thin membrane to **shake and vibrate back and forth thousands of times a second**.

Step 5: Into Your Ears

- As the membrane vibrates, it pushes the **air** next to it, creating invisible sound waves in the air.
- These waves travel into your ear canal and hit your **eardrum**, causing it to vibrate in the exact same rhythm.
- Your brain registers those vibrations as **music, vocals, and beats!**